

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++

ML Frameworks/Tools: PyTorch, TensorFlow, Docker, FastAPI, AWS SageMaker, Kubernetes, Git, Airflow, TensorRT, Hugging Face Transformers

Competence: Machine Learning, Deep Neural Networks, Perception, Segmentation, Prediction, MLOps, Data Science, Computer Vision, CNN, RNN, Transformers, Continuous Integration

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated advanced deep learning models for metrological applications, enhancing model performance and training efficiency by optimizing architectures.
- Explored novel neural network architectures and physics-informed hybrid loss functions to improve model accuracy and stability.
- Collaborated with PhD researchers on metrology-focused ML applications, contributing to a publication at FLOMEKO conference.
- Developed data visualization dashboards for real-time model performance monitoring and analysis.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Achieved 94% vitiligo segmentation accuracy and improved inference speed by 5x using advanced AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and processing time by 6x, demonstrating edge processing capabilities.
- Managed MLOps pipelines on AWS SageMaker, addressing data drift for YOLO perception models and maintaining Kubernetes infrastructure for scalable deployments.
- Led educational sessions on machine learning and big data concepts for university audiences, fostering knowledge transfer.
- Developed Proof-of-Concepts (POCs) for clients utilizing YOLO, few-shot learning, and CNN architectures for diverse perception tasks.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Contributed to full-stack development for AI-based services website using Flask, HTML, and CSS.

- Developed an AI chatbot application leveraging OpenAI API and modern LLM frameworks, showcasing robust AI integration.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Architected and implemented robust web scrapers and ETL pipelines for structured and unstructured data ingestion.
- Automated data ingestion and cleaning workflows, enhancing data quality for client analytics and machine learning applications.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for advanced feature extraction in protein function prediction.
- Increased model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces, demonstrating strong analytical and ML optimization skills.

Razanalytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting model utilizing modern ML techniques for time series prediction and analysis.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to develop the world's first AI-driven self-playing Chitralli Sitar, integrating signal processing and advanced music generation LLMs.
- Achieved significant media recognition, with the project featured on Coke Studio in a music video.

Hugging Face

Open-Source

Contributor

- Co-authored comprehensive technical documentation and implementation for Zero-shot Image Classification on the Hugging Face Hub, enhancing open-source accessibility.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Pursuing advanced coursework in machine learning, deep learning, computer vision, and AI for complex problem-solving.
- Developing strong analytical and mathematical foundations critical for AI/ML engineering roles.

National University of Computer and Emerging Sciences **Aug 2018 – June 2022**
Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor
- Focused on core computer science principles, data structures, algorithms, and machine learning fundamentals.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python